4.3 Permitted Piping Lengths and Level Differences

The piping length and level difference requirements that apply are summarized in Table 3-4.3 and are fully described as follows (refer to Figure 3-4.2):

1. **Requirement 1:** The total length of piping in one refrigerant system should not exceed 1000m. When calculating the total length of piping, the actual length of the indoor main pipes (the piping between the first indoor branch joint and all other indoor branch joints, L_2 to L_{12}) should be doubled.
2. **Requirement 2:** The piping between the farthest indoor unit (N_9) and the first outdoor branch joint (N) should not exceed 175m (actual length) and 200m (equivalent length). (The equivalent length of each branch joint is 0.5m.)
3. **Requirement 3:** The piping between the farthest indoor unit (N_9) and first indoor branch joint (A) should not exceed 40m in length ($\Sigma\{L_7 \text{ to } L_{10}\} + i \le 40$m) unless the following conditions are met and the following measures are taken, in which case the permitted length is up to 90m:

   Conditions:

   a) Each indoor auxiliary pipe (from each indoor unit to its nearest branch joint) joint does not exceed 40m in length (a to m each ≤ 40m).

   b) The difference in length between {the piping from first indoor branch joint (A) to the farthest indoor unit (N_9)} and {the piping from the first indoor branch joint (A) to the nearest indoor unit (N_1)} does not exceed 40m. That is: $(\Sigma\{L_7 \text{ to } L_{10}\} + i) - (\Sigma\{L_2 \text{ to } L_3\} + a) \le 40$m.

   Measures:

   a) Increase the diameter of the indoor main pipes (the piping between the first indoor branch joint and all other indoor branch joints, L_2 to L_{12}) as per Table 3-4.2, except for indoor main pipes which are already the same size as the main pipe (L_1), for which no diameter increases are required.
4. **Requirement 4:** The largest level difference between indoor unit and outdoor unit should not exceed 90m (if the outdoor unit is above) or 110m (if the outdoor unit is below). Additionally: (i) If the outdoor unit is above and the level difference is greater than 20m, it is recommended that an oil return bend with dimensions as specified in Figure 3-4.1 is set every 10m in the gas pipe of the main pipe; and (ii) if the outdoor unit is below and the level difference is more than 40m, the liquid pipe of the main pipe (L_1) should be increased as per Table 3-4.2.
5. **Requirement 5:** The largest level difference between indoor units should not exceed 30m.

Table 3-4.2: Diameter increase requirements

Original (mm)	Increased (mm)
Φ9.53	Φ12.7
Φ12.7	Φ15.9
Φ15.9	Φ19.1
Φ19.1	Φ22.2
Φ22.2	Φ25.4
Φ25.4	Φ28.6
Φ28.6	Φ31.8
Φ31.8	Φ38.1
Φ38.1	Φ41.3
Φ41.3	Φ44.5
Φ44.5	Φ54.0

Figure 3-4.1: Oil return bend (unit: mm)

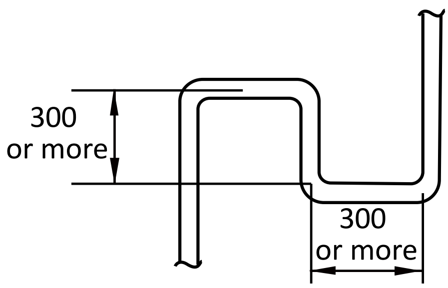
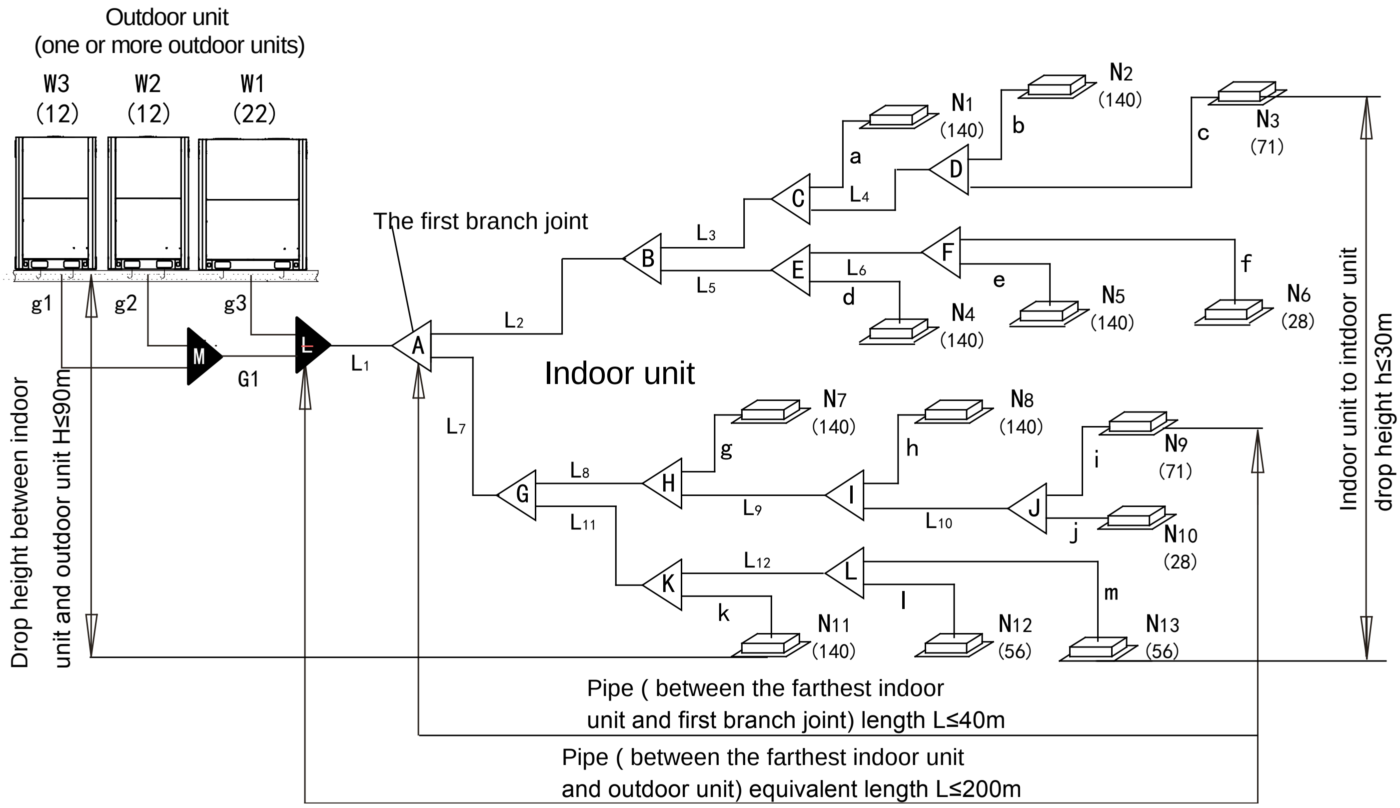


Figure 3-4.2: Permitted refrigerant piping lengths and level differences



Legend		
L_1	Main pipe	Figures in parentheses indicate indoor unit capacity indexes.
L_2 to L_{12}	Indoor main pipes	
a to m	Indoor auxiliary pipes	
A to L	Indoor branch joints	
M, N	Outdoor branch joints	
g1 to g3, G_1	Outdoor connection pipes	

Table 3-4.3: Summary of permitted refrigerant piping lengths and level differences

			Permitted values	Piping in Figure 3-4.2
Piping lengths	**Total piping length**[1]		≤ 1000m	$L_1 + 2 \times \Sigma\{L_2 \text{ to } L_{12}\} + \Sigma\{a \text{ to } m\}$
	Piping between farthest indoor unit and first outdoor branch joint[2]	**Actual length**	≤ 175m	$L_1 + \Sigma\{L_7 \text{ to } L_{10}\} + i$
		Equivalent length	≤ 200m	
	Piping between farthest indoor unit and first indoor branch joint[3]		≤ 40m / 90m	$\Sigma\{L_7 \text{ to } L_{10}\} + i$
Level differences	**Largest level difference between indoor unit and outdoor unit**[4]	**Outdoor unit is above**	≤ 90m	
		Outdoor unit is below	≤ 110m	
	Largest level difference between indoor units[5]		≤ 30m	

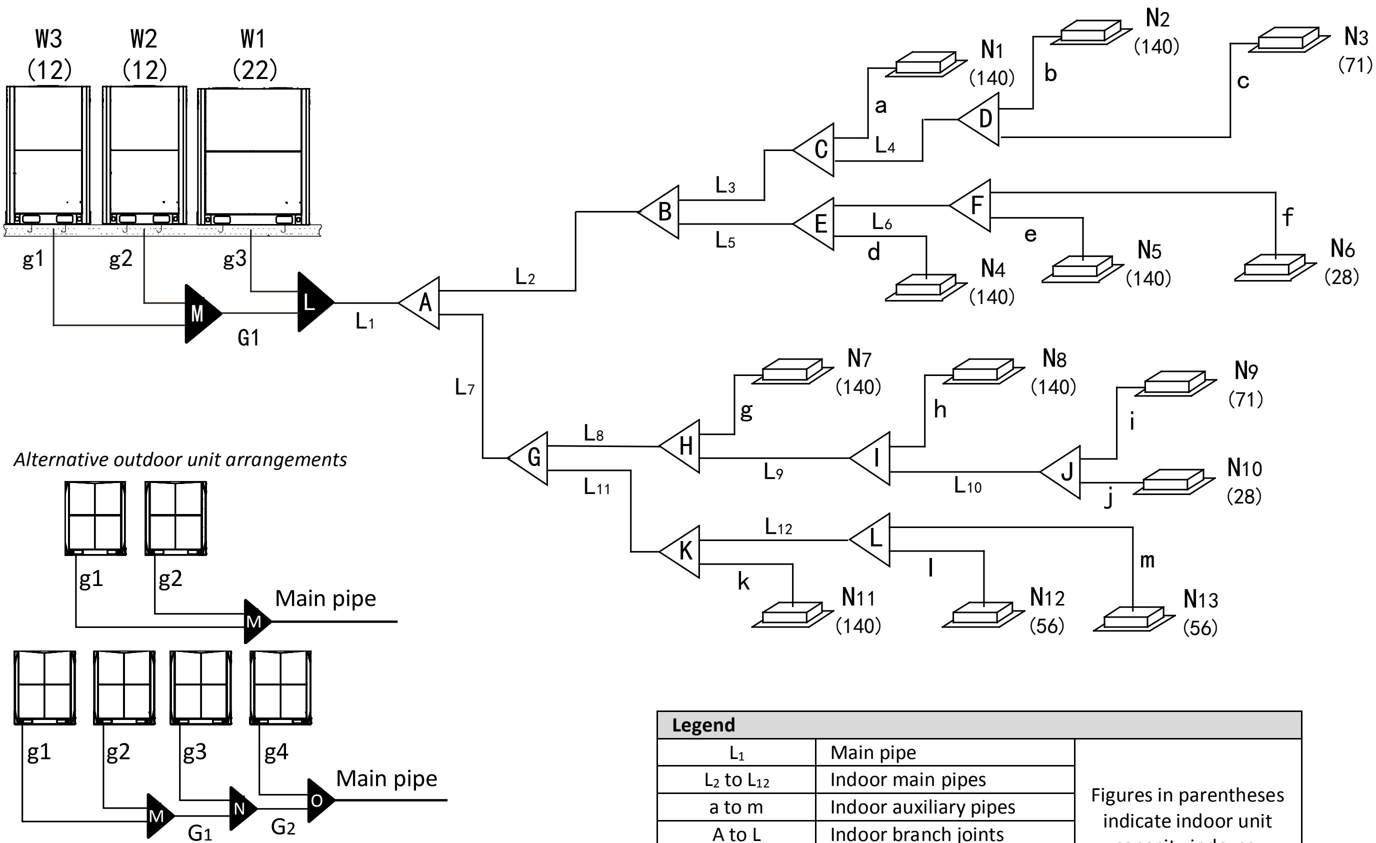
Notes:

1. Refer to Requirement 1, above.
2. Refer to Requirement 2, above.
3. Refer to Requirement 3, above.
4. Refer to Requirement 4, above.
5. Refer to Requirement 5, above.

4.4 Selecting Piping Diameters

Tables 3-4.4 to 3-4.8, below, specify the required pipe diameters for the indoor and outdoor piping. The main pipe (L_1) and first indoor branch joint (A) should be sized according to whichever of Tables 3-4.4 and 3-4.5 indicates the larger size.

Figure 3-4.3: Selecting piping diameters



Legend		
L_1	Main pipe	Figures in parentheses indicate indoor unit capacity indexes.
L_2 to L_{12}	Indoor main pipes	
a to m	Indoor auxiliary pipes	
A to L	Indoor branch joints	
M to O	Outdoor branch joints	
g_1 to g_4, G_1, G_2	Outdoor connection pipes	

Table 3-4.4: Main pipe[1] (L_1), indoor main pipes (L_2 to L_{12}) and indoor branch joint kits

Total capacity of indoor units (kW)	Gas pipe (mm)	Liquid pipe (mm)	Branch joint kit
Capacity < 16.6	Φ15.9	Φ9.53	FQZHN-01D
16.6 ≤ Capacity < 23	Φ19.1	Φ9.53	FQZHN-01D
23 ≤ Capacity < 33	Φ22.2	Φ9.53	FQZHN-02D
33 ≤ Capacity < 46	Φ28.6	Φ12.7	FQZHN-03D
46 ≤ Capacity < 66	Φ28.6	Φ15.9	FQZHN-03D
66 ≤ Capacity < 92	Φ31.8	Φ19.1	FQZHN-03D
92 ≤ Capacity < 135	Φ38.1	Φ19.1	FQZHN-04D
135 ≤ Capacity < 180	Φ41.3	Φ22.2	FQZHN-05D
180 ≤ Capacity	Φ44.5	Φ25.4	FQZHN-05D

Notes:

1. The main pipe (L_1) and first indoor branch joint (A) should be sized according to whichever of Tables 3-4.4 and 3-4.5 indicates the larger size.

Table 3-4.5: Main pipe[1] (L_1) and first indoor branch joint (A)

Total capacity of outdoor units	Equivalent length of all liquid pipes < 90m			Equivalent length of all liquid pipes ≥ 90m		
	Gas pipe (mm)	Liquid pipe (mm)	Branch joint kit	Gas pipe (mm)	Liquid pipe (mm)	Branch joint kit
8HP	Φ22.2	Φ9.53	FQZHN-02D	Φ22.2	Φ12.7	FQZHN-02D
10HP	Φ22.2	Φ9.53	FQZHN-02D	Φ25.4	Φ12.7	FQZHN-02D
12-14HP	Φ25.4	Φ12.7	FQZHN-02D	Φ28.6	Φ15.9	FQZHN-03D
16HP	Φ28.6	Φ12.7	FQZHN-03D	Φ31.8	Φ15.9	FQZHN-03D
18-22HP	Φ28.6	Φ15.9	FQZHN-03D	Φ31.8	Φ19.1	FQZHN-03D
24HP	Φ28.6	Φ15.9	FQZHN-03D	Φ31.8	Φ19.1	FQZHN-03D
26-34HP	Φ31.8	Φ19.1	FQZHN-03D	Φ38.1	Φ22.2	FQZHN-04D
36-50HP	Φ38.1	Φ19.1	FQZHN-04D	Φ38.1	Φ22.2	FQZHN-04D
52-66HP	Φ41.3	Φ22.2	FQZHN-05D	Φ44.5	Φ25.4	FQZHN-05D
68-88HP	Φ44.5	Φ25.4	FQZHN-05D	Φ54.0	Φ25.4	FQZHN-06D

Notes:
1. The main pipe (L_1) and first indoor branch joint (A) should be sized according to whichever of Tables 3-4.4 and 3-4.5 indicates the larger size.

Table 3-4.6: Outdoor connection pipes (g1 to g4, G_1, G_2)

Pipes	Outdoor unit capacity	Gas pipe (mm)	Liquid pipe (mm)
g1 to g4	8-12HP	Φ25.4	Φ12.7
	14-22HP	Φ31.8	Φ15.9
G_1		Φ38.1	Φ19.1
G_2		Φ41.3	Φ22.2

Table 3-4.7: Outdoor branch joint kits (N to O)

No. of outdoor units	Branch joint kit
2	FQZHW-02N1D
3	FQZHW-03N1D
4	FQZHW-04N1D

Table 3-4.8: Indoor auxiliary pipes (a to m)

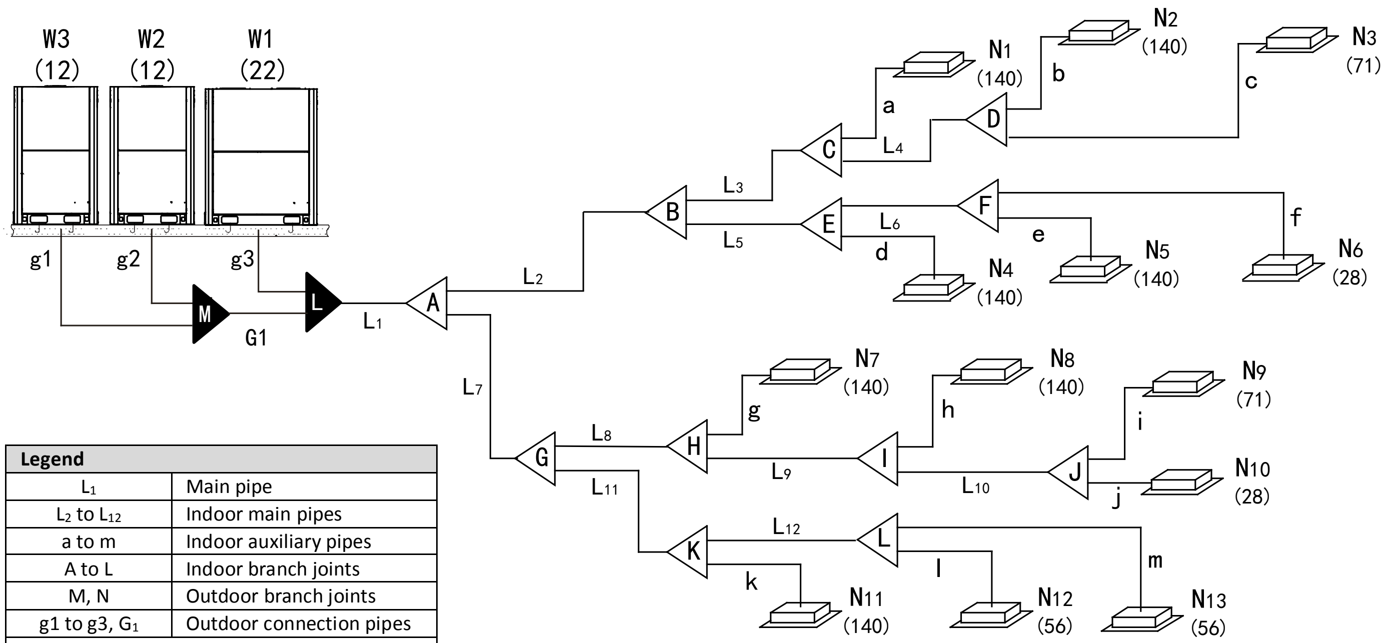
Capacity of indoor unit (kW)	Pipe length ≤ 10m		Pipe length > 10m[1]	
	Gas pipe (mm)	Liquid pipe (mm)	Gas pipe (mm)	Liquid pipe (mm)
≤ 4.5	Φ12.7	Φ6.35	Φ15.9	Φ9.53
≥ 5.6	Φ15.9	Φ9.53	Φ19.1	Φ12.7

Notes:
1. An indoor auxiliary pipe should not be larger than the indoor main pipe immediately upstream of it. For indoor auxiliary pipes greater than 10m in length with indoor units of capacity great than or equal to 5.6kW, the gas and liquid side pipes should each either be sized according to this table, or else be the same size as the indoor main pipe immediately upstream, whichever is smaller.

4.5 Refrigerant Piping Selection Example

The example below illustrates the piping selection procedure for a system consisting of three outdoor units (22HP + 12HP + 12HP) and 13 indoor units. The system's equivalent total piping length is in excess of 90m; the piping between the farthest indoor unit and the first indoor branch joint is less than 40m in length; and each indoor auxiliary pipe (from each indoor unit to its nearest branch joint) is less than 10m in length.

Figure 3-4.4: Refrigerant piping selection example



Legend	
L_1	Main pipe
L_2 to L_{12}	Indoor main pipes
a to m	Indoor auxiliary pipes
A to L	Indoor branch joints
M, N	Outdoor branch joints
g1 to g3, G_1	Outdoor connection pipes
Figures in parentheses indicate indoor unit capacity indexes.	

Step 1: Select indoor auxiliary pipes

- Indoor units N_1 to N_5, N_7 to N_9 and N_{11} to N_{13} are of capacity 5.6kW or greater and their indoor auxiliary pipes are less than 10m in length. Refer to Table 3-4.8. Indoor auxiliary pipes a to e, g to i, and k to m are Φ15.9 / Φ9.53.
- Indoor units N_6 and N_{10} are of capacity less than 4.5kW and their indoor auxiliary pipes are less than 10m in length. Refer to Table 3-4.8. Indoor auxiliary pipes f and j are Φ12.7 / Φ6.35.

Step 2: Select indoor main pipes and indoor branch joints B to L

- The indoor units (N_2 and N_3) downstream of indoor branch joint D have total capacity of 14 + 7.1 = 21.1kW. Refer to Table 3-4.4. Indoor main pipe L_4 is Φ19.1 / Φ9.53. Indoor branch joint D is FQZHN-01D.
- The indoor units (N_1 to N_6) downstream of indoor branch joint B have total capacity of 14 x 4 + 7.1 + 2.8 = 65.9kW. Refer to Table 3-4.4. Indoor main pipe L_2 is Φ28.6 / Φ15.9. Indoor branch joint B is FQZHN-03D.
- The other indoor main pipes and indoor branch joints C and E to L are selected in the same fashion.

Step 3: Select main pipe and indoor branch joint A

- The indoor units (N_1 to N_{13}) downstream of indoor branch joint A have total capacity of 14 x 7 + 7.1 x 2 + 5.6 x 2 + 2.8 x 2 = 129kW. The system's equivalent total piping length is in excess of 90m. The total capacity of the outdoor units is 22 + 12 + 12 = 46HP. Refer to Tables 3-4.4 and 3-4.5. Main pipe L_1 is the larger of Φ38.1 / Φ19.1 and Φ38.1 / Φ22.2, hence Φ38.1 / Φ22.2. Indoor branch joint A is FQZHN-04D.

Step 4: Select outdoor connection pipes and outdoor branch joints

- The master unit is 22HP and the slave units are 12HP. Refer to Table 3-4.6. Outdoor connection pipes g1 and g2 are Φ25.4 / Φ12.7 and outdoor connection pipe g3 is Φ31.8 / Φ15.9.
- Refer to Table 3-4.6. Outdoor connection pipe G_1 is Φ38.1 / Φ19.1.
- There are three outdoor units in the system. Refer to Table 3-4.7. Outdoor branch joints M and N are FQZHW-03N1D.

4.6 Branch Joints

Branch joint design should take account of the following:

- U-shaped branch joints should be used – tee joints are not suitable. Branch joint dimensions are given in Tables 3-4.9 and 3-4.10.
- To avoid accumulation of oil in the outdoor units, outdoor branch joints should be installed horizontally and must not be higher than the outdoor unit refrigerant outlets. Refer to Figure 3-5.9 in Part 3, 5.6 "Branch Joints". Indoor branch joints may be installed either horizontally or vertically.
- To ensure even distribution of refrigerant, branch joints should not be installed within 500mm of a 90° bend, another branch joint or the straight section of piping leading to an indoor unit, with the minimum 500mm being measured from the point where the branch joint is connected to the piping, as shown in Figure 3-4.5.

Figure 3-4.5: Branch joint spacing and separation from bends (unit: mm)